

WORKED EXAMPLE

Drop hidden activations with a fixed binary mask

The course convention is standard dropout: train with $M \cdot h$; infer with $q \cdot h$.

A mask entry is 1 when its hidden unit is kept and 0 when dropped. Each entry is an independent Bernoulli draw: it equals 1 with keep probability q , called alpha in the lecture.

$$q = 0.5, \quad M = (1, 0), \quad x = 1, \quad t = 1, \quad \eta = 0.01$$

Only the two hidden activations are eligible for dropout in this exercise. The input, linear output and bias constants are not masked.

Cache the sampled mask for the entire forward/backward pass. A later example or update can receive a new mask under the chosen sampling policy.

WORKED EXAMPLE

Define all seven trainable parameters

Two ReLU hidden units feed one linear output.

$$z_j = a_j x + c_j, \quad h_j = \max(0, z_j), \quad d_j = M_j h_j$$

$$\hat{y} = u_1 d_1 + u_2 d_2 + b, \quad L = \frac{1}{2}(\hat{y} - t)^2$$

$$(a_1, a_2, c_1, c_2, u_1, u_2, b) = (1, 2, 1, -1, 3, -2, 0.5)$$

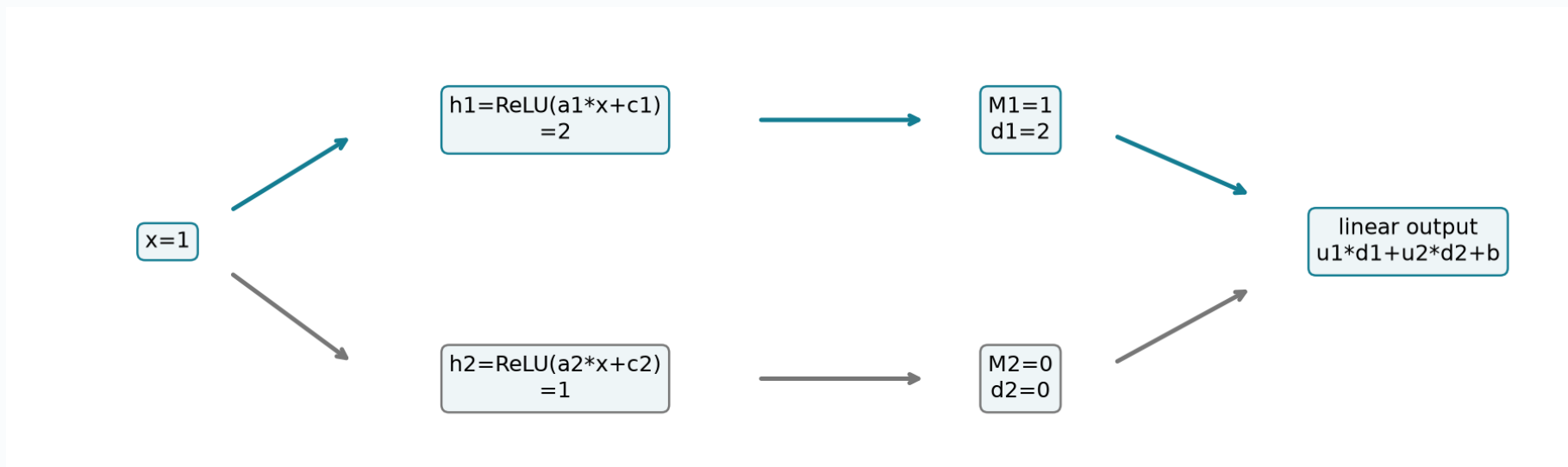
a_1 and a_2 are input weights; c_1 and c_2 are hidden biases; u_1 and u_2 are output weights; b is the output bias. Count each bias once.

The lecture uses general layer notation. This small network makes its mask and chain-rule equations explicit. There is no additional L2 penalty in this lesson.

WORKED EXAMPLE

Follow the two hidden paths

The second hidden unit is computed, then its activation is multiplied by zero.



A dropped path sends zero activation to the output and zero data-loss derivative backward through the mask. Its parameters remain stored; they are not deleted.

WORKED EXAMPLE

Main pass: hidden activations

Use the original parameters for this separate pass.

$$z_1 = 1(1) + 1 = 2, \quad z_2 = 2(1) + (-1) = 1$$

$$h_1 = \max(0, 2) = 2, \quad h_2 = \max(0, 1) = 1$$

$$M = (1, 0), \quad q = 0.5$$

Both preactivations are positive, so each ReLU local derivative is 1. A zero dropout mask is different from a ReLU being inactive because of a nonpositive preactivation.

Standard dropout multiplies the hidden activations by the fixed mask with no training-time division by q .

WORKED EXAMPLE

Main pass: mask, output and loss

Use the masked activations in the output weighted sum.

$$d_1 = 1(2) = 2, \quad d_2 = 0(1) = 0$$

$$\hat{y} = 3(2) + (-2)(0) + 0.5 = 6.5$$

$$r = \hat{y} - t = 6.5 - 1 = 5.5$$

$$L = \frac{1}{2}(5.5)^2 = 15.125$$

The output is linear, so its derivative with respect to the weighted sum is 1. The output bias is added once and is not scaled by the hidden mask.

WORKED EXAMPLE

Main pass: output gradients

Derivative of the half-square with respect to prediction is the residual.

$$\frac{\partial L}{\partial \hat{y}} = r, \quad \frac{\partial L}{\partial u_j} = r d_j, \quad \frac{\partial L}{\partial b} = r$$

$$g_{u_1} = 5.5(2) = 11$$

$$g_{u_2} = 5.5(0) = 0, \quad g_b = 5.5$$

The outgoing weight of a dropped unit has zero data gradient because its source activation d is zero. The output bias can still have a nonzero gradient.

WORKED EXAMPLE

Main pass: hidden deltas

Multiply the backward derivative through the same cached mask and the local ReLU slope.

$$\delta_j = \frac{\partial L}{\partial z_j} = ru_j \frac{\partial d_j}{\partial h_j} \text{ReLU}'(z_j)$$

$$\partial d_j / \partial h_j = M_j$$

$$\delta_1 = 5.5(3)(1)(1) = 16.5$$

$$\delta_2 = 5.5(-2)(0)(1) = 0$$

Use original output weights 3 and -2, not values updated from the preceding gradient calculation.
No parameter has been changed during this backward pass.

WORKED EXAMPLE

Main pass: input weights and hidden biases

An input weight contributes x ; a hidden bias contributes 1.

$$\frac{\partial L}{\partial a_j} = \delta_j x, \quad \frac{\partial L}{\partial c_j} = \delta_j$$

$$g_{a_1} = 16.5(1) = 16.5, \quad g_{c_1} = 16.5$$

$$g_{a_2} = 0(1) = 0, \quad g_{c_2} = 0$$

A dropped hidden bias has zero data gradient through the masked path. This does not mean that the bias constant itself was directly masked.

These are data-loss gradients. If a separate regularizer penalized a dropped unit's weights, it could still contribute nonzero gradients.

WORKED EXAMPLE

Main pass: update hidden parameters

Subtract eta times each gradient; unchanged zero-gradient parameters remain stored.

$$(a_1)_{new} = 1 - 0.01(16.5) = 0.835$$

$$(a_2)_{new} = 2 - 0.01(0) = 2$$

$$(c_1)_{new} = 1 - 0.01(16.5) = 0.835$$

$$(c_2)_{new} = -1 - 0.01(0) = -1$$

All seven new values use the gradients from the original common state. Do not recompute later gradients using already updated parameters.

WORKED EXAMPLE

Main pass: update output parameters

Subtract eta times each gradient; unchanged zero-gradient parameters remain stored.

$$(u_1)_{new} = 3 - 0.01(11) = 2.89$$

$$(u_2)_{new} = -2 - 0.01(0) = -2$$

$$(b)_{new} = 0.5 - 0.01(5.5) = 0.445$$

All seven new values use the gradients from the original common state. Do not recompute later gradients using already updated parameters.

WORKED EXAMPLE

Main pass: check the updated pass

Diagnostic only: reuse the cached mask to evaluate the same loss function.

$$z^{new} = (0.835 + 0.835, 2 + (-1)) = (1.67, 1)$$

$$d^{new} = (1.67, 0)$$

$$\hat{y}^{new} = 2.89(1.67) + (-2)(0) + 0.445 = 5.2713$$

$$L^{new} = \frac{1}{2}(4.2713)^2 = 9.122001845$$

This same-mask diagnostic is not a requirement to reuse this mask forever. Later training examples or updates sample new masks under the declared policy.

WORKED EXAMPLE

Standard-dropout inference: reset original parameters

Use the original $h=(2,1)$, $u=(3,-2)$, $b=.5$ for this comparison.

$$\mathbb{E}[M_j h_j] = q h_j \quad \text{for fixed } h_j$$

$$d_{test} = qh = (0.5(2), 0.5(1)) = (1, 0.5)$$

$$\hat{y}_{test} = 3(1) + (-2)(0.5) + 0.5 = 2.5$$

$$\text{equivalently : } (qu) = (1.5, -1)$$

$$1.5(2) + (-1)(1) + 0.5 = 2.5$$

Scale the activations OR their outgoing weights, not both. The next additive output bias stays .5.

No random mask is sampled for this standard deterministic inference rule.

WORKED EXAMPLE

Inverted comparison: hidden activations

Use the original parameters for this separate pass.

$$z_1 = 1(1) + 1 = 2, \quad z_2 = 2(1) + (-1) = 1$$

$$h_1 = \max(0, 2) = 2, \quad h_2 = \max(0, 1) = 1$$

$$M = (1, 0), \quad q = 0.5$$

Both preactivations are positive, so each ReLU local derivative is 1. A zero dropout mask is different from a ReLU being inactive because of a nonpositive preactivation.

This is an inverted-dropout comparison, reset to original parameters; retained activations divide by q during training.

WORKED EXAMPLE

Inverted comparison: mask, output and loss

Use the masked activations in the output weighted sum.

$$d_1 = 1(2)/0.5 = 4, \quad d_2 = 0(1)/0.5 = 0$$

$$\hat{y} = 3(4) + (-2)(0) + 0.5 = 12.5$$

$$r = \hat{y} - t = 12.5 - 1 = 11.5$$

$$L = \frac{1}{2}(11.5)^2 = 66.125$$

The output is linear, so its derivative with respect to the weighted sum is 1. The output bias is added once and is not scaled by the hidden mask.

WORKED EXAMPLE

Inverted comparison: output gradients

Derivative of the half-square with respect to prediction is the residual.

$$\frac{\partial L}{\partial \hat{y}} = r, \quad \frac{\partial L}{\partial u_j} = r d_j, \quad \frac{\partial L}{\partial b} = r$$

$$g_{u_1} = 11.5(4) = 46$$

$$g_{u_2} = 11.5(0) = 0, \quad g_b = 11.5$$

The outgoing weight of a dropped unit has zero data gradient because its source activation d is zero. The output bias can still have a nonzero gradient.

WORKED EXAMPLE

Inverted comparison: hidden deltas

Multiply the backward derivative through the same cached mask and the local ReLU slope.

$$\delta_j = \frac{\partial L}{\partial z_j} = ru_j \frac{\partial d_j}{\partial h_j} \text{ReLU}'(z_j)$$

$$\partial d_j / \partial h_j = M_j / q$$

$$\delta_1 = 11.5(3)(2)(1) = 69$$

$$\delta_2 = 11.5(-2)(0)(1) = 0$$

Use original output weights 3 and -2, not values updated from the preceding gradient calculation.
No parameter has been changed during this backward pass.

WORKED EXAMPLE

Inverted comparison: input weights and hidden biases

An input weight contributes x ; a hidden bias contributes 1.

$$\frac{\partial L}{\partial a_j} = \delta_j x, \quad \frac{\partial L}{\partial c_j} = \delta_j$$

$$g_{a_1} = 69(1) = 69, \quad g_{c_1} = 69$$

$$g_{a_2} = 0(1) = 0, \quad g_{c_2} = 0$$

A dropped hidden bias has zero data gradient through the masked path. This does not mean that the bias constant itself was directly masked.

These are data-loss gradients. If a separate regularizer penalized a dropped unit's weights, it could still contribute nonzero gradients.

WORKED EXAMPLE

Inverted-dropout inference: original parameters

Training uses $M \cdot h / q$; inference uses the unscaled h .

$$\mathbb{E}[M_j h_j / q] = h_j$$

$$\hat{y}_{test} = 3(2) + (-2)(1) + 0.5 = 4.5$$

convention	training activation	inference activation
standard	$M \cdot h$	$q \cdot h$
inverted	$M \cdot h / q$	h

The standard result 2.5 and inverted result 4.5 use the same fixed original parameters under different conventions. They are not two separately trained models whose parameters or outputs must match.

Choose one convention and keep its training, backward and inference scaling consistent.

WORKED EXAMPLE

Enumerate all masks at the original state

Two independent mask entries with $q=.5$ give four equally likely masks.

M	probability	active units	standard output
(0,0)	.25	0	.5
(0,1)	.25	1	-1.5
(1,0)	.25	1	6.5
(1,1)	.25	2	4.5

$$\mathbb{E}[\text{active}] = 0.25(0 + 1 + 1 + 2) = 1$$

$$\mathbb{E}[\hat{y}] = 0.25(0.5 - 1.5 + 6.5 + 4.5) = 2.5$$

The mean output equals standard inference exactly here because the output is linear in the masked, fixed hidden activations. These masks share parameters; four masks do not mean four independently trained models.

WORKED EXAMPLE

Nonlinear output: expectation is only approximated

The lecture page 155 uses an approximation for a general network.

$$Z = 2M, \quad P(M = 0) = P(M = 1) = 0.5$$

$$\mathbb{E}[\sigma(Z)] = 0.5\sigma(0) + 0.5\sigma(2)$$

$$= 0.5(0.5) + 0.5(0.880797) \approx 0.690399$$

$$\mathbb{E}[Z] = 0.5(0) + 0.5(2) = 1$$

$$\sigma(\mathbb{E}[Z]) = \sigma(1) \approx 0.731059$$

A function of the mean is not generally the mean of the function. Replacing random activations by their expectations is not an exact ensemble-output identity for arbitrary nonlinear networks.

INDEPENDENT PRACTICE

Your turn: keep the other hidden unit

Reset to the original seven parameters; use standard dropout.

$$M = (0, 1), \quad q = 0.5, \quad x = 1, \quad t = 1, \quad \eta = 0.01$$

Compute the hidden activations, masked values, prediction, loss, all seven gradients and all seven simultaneous updates. Recompute the loss with the same mask as a diagnostic.

Separately, consider three independently masked eligible units with keep probability .8. Find the number of possible masks, expected active count and probability of the particular ordered mask (1,1,0).

WORKED ANSWER

Practice: hidden activations

Use the original parameters for this separate pass.

$$z_1 = 1(1) + 1 = 2, \quad z_2 = 2(1) + (-1) = 1$$

$$h_1 = \max(0, 2) = 2, \quad h_2 = \max(0, 1) = 1$$

$$M = (0, 1), \quad q = 0.5$$

Both preactivations are positive, so each ReLU local derivative is 1. A zero dropout mask is different from a ReLU being inactive because of a nonpositive preactivation.

Standard dropout multiplies the hidden activations by the fixed mask with no training-time division by q .

WORKED ANSWER

Practice: mask, output and loss

Use the masked activations in the output weighted sum.

$$d_1 = 0(2) = 0, \quad d_2 = 1(1) = 1$$

$$\hat{y} = 3(0) + (-2)(1) + 0.5 = -1.5$$

$$r = \hat{y} - t = -1.5 - 1 = -2.5$$

$$L = \frac{1}{2}(-2.5)^2 = 3.125$$

The output is linear, so its derivative with respect to the weighted sum is 1. The output bias is added once and is not scaled by the hidden mask.

WORKED ANSWER

Practice: output gradients

Derivative of the half-square with respect to prediction is the residual.

$$\frac{\partial L}{\partial \hat{y}} = r, \quad \frac{\partial L}{\partial u_j} = r d_j, \quad \frac{\partial L}{\partial b} = r$$

$$g_{u_1} = -2.5(0) = 0$$

$$g_{u_2} = -2.5(1) = -2.5, \quad g_b = -2.5$$

The outgoing weight of a dropped unit has zero data gradient because its source activation d is zero. The output bias can still have a nonzero gradient.

WORKED ANSWER

Practice: hidden deltas

Multiply the backward derivative through the same cached mask and the local ReLU slope.

$$\delta_j = \frac{\partial L}{\partial z_j} = ru_j \frac{\partial d_j}{\partial h_j} \text{ReLU}'(z_j)$$

$$\partial d_j / \partial h_j = M_j$$

$$\delta_1 = -2.5(3)(0)(1) = 0$$

$$\delta_2 = -2.5(-2)(1)(1) = 5$$

Use original output weights 3 and -2, not values updated from the preceding gradient calculation.
No parameter has been changed during this backward pass.

WORKED ANSWER

Practice: input weights and hidden biases

An input weight contributes x ; a hidden bias contributes 1.

$$\frac{\partial L}{\partial a_j} = \delta_j x, \quad \frac{\partial L}{\partial c_j} = \delta_j$$

$$g_{a_1} = 0(1) = 0, \quad g_{c_1} = 0$$

$$g_{a_2} = 5(1) = 5, \quad g_{c_2} = 5$$

A dropped hidden bias has zero data gradient through the masked path. This does not mean that the bias constant itself was directly masked.

These are data-loss gradients. If a separate regularizer penalized a dropped unit's weights, it could still contribute nonzero gradients.

WORKED ANSWER

Practice: update hidden parameters

Subtract eta times each gradient; unchanged zero-gradient parameters remain stored.

$$(a_1)_{new} = 1 - 0.01(0) = 1$$

$$(a_2)_{new} = 2 - 0.01(5) = 1.95$$

$$(c_1)_{new} = 1 - 0.01(0) = 1$$

$$(c_2)_{new} = -1 - 0.01(5) = -1.05$$

All seven new values use the gradients from the original common state. Do not recompute later gradients using already updated parameters.

WORKED ANSWER

Practice: update output parameters

Subtract eta times each gradient; unchanged zero-gradient parameters remain stored.

$$(u_1)_{new} = 3 - 0.01(0) = 3$$

$$(u_2)_{new} = -2 - 0.01(-2.5) = -1.975$$

$$(b)_{new} = 0.5 - 0.01(-2.5) = 0.525$$

All seven new values use the gradients from the original common state. Do not recompute later gradients using already updated parameters.

WORKED ANSWER

Practice: check the updated pass

Diagnostic only: reuse the cached mask to evaluate the same loss function.

$$z^{new} = (1 + 1, 1.95 + (-1.05)) = (2, 0.9)$$

$$d^{new} = (0, 0.9)$$

$$\hat{y}^{new} = 3(0) + (-1.975)(0.9) + 0.525 = -1.2525$$

$$L^{new} = \frac{1}{2}(-2.2525)^2 = 2.536878125$$

This same-mask diagnostic is not a requirement to reuse this mask forever. Later training examples or updates sample new masks under the declared policy.

WORKED ANSWER

Answer: mask count and probability

Independent entries can each be kept or dropped.

$$\text{possible masks} = 2^3 = 8$$

$$\mathbb{E}[\text{active}] = 0.8 + 0.8 + 0.8 = 2.4$$

$$P(M = (1, 1, 0)) = 0.8(0.8)(1 - 0.8) = 0.128$$

The expected count 2.4 is an average across masks; each actual count is an integer. The eight masks are not equally probable because q is not .5.

There is no combinatorial factor for one specified ordered mask. If asked for any mask with exactly two active units, three different masks qualify, so that probability would be $3(.128) = .384$.

Before an exam calculation, name the dropout convention, cache the mask, keep original weights during backpropagation and scale only eligible activations.